Yikai Tang

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Education

Shanghai Jiao Tong University (SJTU) *Sep 2022 – Present*
Advisor: **Prof. Yunbo Wang** B.E. in Computer Science and Technology (IEEE Honor Class)

- **GPA: 3.8**
- **27** A/A+ Courses: Design and Analysis of Algorithms,Operating System,Discrete Mathematics, etc.
- GRE: V160, Q168; TOEFL: 110

University of California, Berkeley *Jan 2025 – May 2025*
Advisor: **Prof. Jitendra Malik** Visiting Student, BGA Program

- CS 184: Foundations of Computer Graphics — **A**
- CS 61A: Structure and Interpretation of Computer Programs — **A**
- EECS 126: Probability and Random Processes — **A-**

Publications

DIPOLE *BAIR, UC Berkeley* | *Mar 2025 – Sep 2025*
Fusing Vision and Geometry for Robust Visuomotor Generalization
Y. Tang*, *H. Geng*, J. Jia, Y. Hu, S. Zang, J. Yang, P. Abbeel, J. Malik*

- Proposed a **modality-aware dropout mechanism** enabling robust RGB–point-cloud fusion within diffusion policies.
- Built a unified visual–geometric latent space that improves generalization across lighting, materials, viewpoints, and object configurations.
- Demonstrated strong, robust, state-of-the-art performance through geometry-informed policy learning.

ISF-VLA **IROS 2026 (Accepted)** | *BAIR, UC Berkeley* | *Sep 2025 – Nov 2025*
An Implicit 3D Vision-Language-Action Model Capitalizing on 3D-Aware Vision Foundation Models through Implicit Spatial Fusion
S. Zang, S. Wei*, H. Geng,* ***Y. Tang****, B. An, S. Jayavelu, P. Abbeel, J. Malik*

- Introduced **implicit 3D spatial fusion** to integrate geometric cues into RGB–point-cloud fusion **for VLA**.
- Achieved significant gains in cross-scene and cross-viewpoint robustness through 3D-grounded multimodal representation learning.
- Enabled stable and expressive fusion for large pretrained visual–language backbones in manipulation.

RoST3R *BAIR, UC Berkeley* | *Mar 2025 – Sep 2025*
Robot-Aware Dynamic 3D Reconstruction from RGB Input for Robotic Manipulation
Y. Han, H. Geng, J. Zhang, Z. Chen, ***Y. Tang****, C. Cheng, P. Li, T. Darrell, P. Abbeel, J. Malik, Y. Wang*

- Developed a dynamic, metric-scale 3D reconstruction pipeline from monocular RGB tailored for robotic manipulation.
- Enabled consistent 3D scene understanding with temporally stable geometry beneficial for downstream control.
- Delivered robot-aware reconstruction that integrates motion cues for improved dynamic-scene fidelity.

Research Experience

BAIR, UC Berkeley *Mar 2025 – Nov 2025*
Advisor: **Prof. Jitendra Malik**

DIPOLE* — Multimodal diffusion policy with modality-aware fusion *Mar 2025 – Sep 2025*
RoST3R — Robot-aware dynamic 3D reconstruction from RGB *Mar 2025 – Sep 2025*
ISF-VLA — Implicit 3D spatial fusion for VLA models *Sep 2025 – Nov 2025*

Yunbo Lab, SJTU *Oct 2023 – Dec 2024*
Advisor: **Prof. Yunbo Wang**

DeformSDF* — Physics-aware SDF reconstruction for highly deformable objects *Aug 2024 – Dec 2024*

Intuitive Physics Fluids — Multi-view real-world fluid reconstruction *Oct 2023 – Jul 2024*

*First / Co-First Author

Projects

Game Design with olcPixelGameEngine *Jun 2023 – Jul 2023*

Advisor: Yuting Wang *Tech: C++*

- Designed and implemented an SLG game with custom UI and unit systems.
- Implemented pixel-level visual effects for movement, attacks, and explosions.

News Crawler and Search Engine *Dec 2023 – Jan 2024*

Advisor: Ya Zhang *Tech: Flask, BeautifulSoup, Lucene, Face_Recognition*

- Built a multithreaded crawler collecting 10k+ news articles and images.
- Implemented a Lucene-based search backend with a Flask web interface.

Structured Macro Support for WhileDB *Dec 2024 – Jan 2025*

Advisor: Qinxiang Cao *Tech: Flex, Bison, C#*

- Extended WhileDB with structured macros and function calls while preserving AST integrity.

Honors and Awards

SJTU Merit Scholarship for Academic Excellence (Category C) — 2023
Class Student Leader in Academics — 2022

Professional Service

Workshop Organizer: **[NeurIPS 2026]** Embodied World Models for Decision Making

Leadership and Activities

Debate Team, SJTU — Competed in 30+ tournaments; mentored 20+ members.

Technical Skills

Python, C++, Coq; PyTorch, ROS2, Transformers, Open3D; Linux, Git